Project: Worked examples
Title: U-values and room heat loss
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the U-values of built-up elements and the steady-state design heat loss of one room. It gives the fabric and ventilation loss, the room load for sizing emitters, the heat loss coefficient and an optional check against target U-values.

WHAT IT CALCULATES	RELATION
Element U-value	$U = 1/(R_{si} + \Sigma\, d/\lambda + R_g + R_{se})$, or the user's U override for ground floors and glazing, or the ground floor record U_g for a floor without layers or an override
Element fabric loss	$\Phi_i = A \cdot U \cdot f \cdot (T_{int} - \theta_{adj})$, with $\theta_{adj} = \theta_e$ where the element faces outside
Thermal bridges	$\Phi_{tb} = y_{tb} \cdot \Sigma(A \cdot f)_ext \cdot (T_{int} - \theta_e)$ on the elements facing outside
Fabric total	$\Phi_T = \Sigma\Phi_i + \Phi_{tb}$
Ventilation loss	$q_v = n \cdot V$; $\Phi_V = 0.34 \cdot q_v \cdot (T_{int} - \theta_e)$, or $(T_{int} - T_{supply})$ through an MVHR supply, plus $0.34 \cdot n_{inf} \cdot V \cdot (T_{int} - \theta_e)$ infiltration with MVHR
Room load	$\Phi_{HL} = \Phi_T + \Phi_V$ (CIBSE DHDG method: × (1 + additional allowance))
Heat loss coefficient	$HLC = \Phi_{HL}/(T_{int} - \theta_e)$; loss per floor area Φ_{HL}/A_{floor}
U-value check	U against the target U of each element (the global target applies to built-up elements facing outside only; an element's own target applies wherever it faces)

Basis: Thermal resistance and transmittance of building components (clause numbers to be checked)[1]; Heating systems in buildings — design heat load: transmission heat loss[2](cl. 6.3.2 (clause number to be checked against the owner's copy)); Heating systems in buildings — design heat load: ventilation heat loss[2](cl. 6.3.3 (clause number to be checked against the owner's copy)); Room heat loss and the additional allowance, reduced method (name and edition unverified)[3].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Calculation method (changes the citations; the DHDG option adds the allowance below)	BS EN 12831-1 elemental		assumption: Elemental method; the design-team default.	any
elements	Building elements bounding the room	5 row(s)		drawing: A-101 rev C	any
layers	Construction layers (several elements share the table, keyed by element name)	8 row(s)		drawing: A-210 rev C	any
V	Room volume	38.88	m³	drawing: A-101 rev C	5 … 1000 m³

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A_{floor}	Floor area, for the loss per floor area (optional; blank = no loss per floor area)	16.2	m²	drawing: A-101 rev C	1 … 200 m²
T_{int}	Internal design temperature	21	°C	spec: SPEC-A-02 §3.1: living-room design temperature	16 … 23 °C
θ_e	External design temperature (location record)	-3	°C	assumption: Illustrative location design temperature; take the project's own from its design-temperature record.	-15 … 3 °C
y_{tb}	Thermal-bridge allowance on the area facing outside	0.1	W/(m²·K)	assumption: Illustrative thermal-bridge allowance; replace with the project's bridging record.	0 … 0.3 W/(m²·K)
f_{add}	Additional allowance on the room load (CIBSE DHDG method only)	0	%	default	0 … 20 %
U_{target}	Target U for built-up elements without their own target (optional; blank = no global target)	0.3	W/(m²·K)	spec: SPEC-A-02 §4.2: external wall U-value target	0.05 … 8 W/(m²·K)
n	Air change rate	0.8	1/h	assumption: Natural ventilation allowance for a living room.	0.1 … 2.5 1/h

TABLE ROW	VALUES
elements[1]	name = Front wall; construction = Cavity wall; area = 8.7 m²
elements[2]	name = Window; area = 2.1 m²; U_user = 1.4 W/(m²·K)
elements[3]	name = Gable wall; construction = Cavity wall; area = 8.64 m²
elements[4]	name = Party wall; area = 8.64 m²; theta_adj = 18 °C; Rg = 0.18 m²·K/W; U_max = 1.2 W/(m²·K)
elements[5]	name = Ground floor; area = 16.2 m²; U_user = 0.25 W/(m²·K)
layers[1]	element = Cavity wall; name = Outer brick leaf; d = 102.5 mm; lambda = 0.84 W/(m·K)
layers[2]	element = Cavity wall; name = Insulation; d = 100 mm; lambda = 0.03 W/(m·K)
layers[3]	element = Cavity wall; name = Inner block leaf; d = 100 mm; lambda = 0.2 W/(m·K)
layers[4]	element = Cavity wall; name = Plaster; d = 13 mm; lambda = 0.4 W/(m·K)
layers[5]	element = Party wall; name = Plaster; d = 13 mm; lambda = 0.4 W/(m·K)
layers[6]	element = Party wall; name = Block leaf 1; d = 100 mm; lambda = 0.6 W/(m·K)

TABLE ROW	VALUES
layers[7]	element = Party wall; name = Block leaf 2; d = 100 mm; lambda = 0.6 W/(m·K)
layers[8]	element = Party wall; name = Plaster; d = 13 mm; lambda = 0.4 W/(m·K)

3. Assumptions

A1. Steady state: every temperature is constant, with no solar or internal gains, no thermal mass and no dynamic simulation. (Design heat loss, not an annual energy estimate.)
A2. Layers are homogeneous and in series, with one-dimensional heat flow. (BS EN ISO 6946:2017 homogeneous-layer calculation.)
A3. Bridged layers such as timber studs, mortar joints and fixings are not corrected for. (BS EN ISO 6946:2017 homogeneous-layer calculation.)
A4. A thermal-bridge allowance y_{tb} is added instead of correcting bridged layers. (Stated.)
A5. A blank Rsi is the conventional BS EN ISO 6946 value for the element's heat-flow direction. (Small conventional BS EN ISO 6946 set (three values), not a table of design data.)
A6. The conventional Rsi is 0.13 m²K/W horizontal, 0.10 upward and 0.17 downward. (Small conventional BS EN ISO 6946 set (three values), not a table of design data.)
A7. A blank Rse is 0.04 m²K/W for an element facing outside. (Small conventional BS EN ISO 6946 set, not a table of design data.)
A8. Where an element faces another space, a blank Rse is the same as Rsi. (Small conventional BS EN ISO 6946 set, not a table of design data.)
A9. The global target U applies to built-up elements facing outside only. (Scope of the target check: an internal partition or floor is checked only against a target entered on its own row.)
A10. Ground-floor and glazing U-values are entered by the user as an override. (Stated limit.)
A11. A ground floor U-value record fills U_g for a floor element without layers or an override. (Stated limit: the record's value is the user's, from its table.)
A12. The calculator does not model ground heat transfer. (Stated limit.)
A13. The air constant 0.34 Wh/(m³·K) is the conventional rounded $\rho \cdot c_p$ of air. (Derived: about 1.2 kg/m³ × 1005 J/(kg·K) ÷ 3600, not read from a table.)
A14. Design temperatures, conductivities, surface resistances and the thermal-bridge allowance are the user's own records. (No table from a standard or a manufacturer is shipped.)
A15. method (Calculation method (changes the citations; the DHDG option adds the allowance below)) is taken as BS EN 12831-1 elemental. (Elemental method; the design-team default.)
A16. θ_e (External design temperature (location record)) is taken as -3 °C. (Illustrative location design temperature; take the project's own from its design-temperature record.)
A17. y_{tb} (Thermal-bridge allowance on the area facing outside) is taken as 0.1 W/(m²·K). (Illustrative thermal-bridge allowance; replace with the project's bridging record.)
A18. n (Air change rate) is taken as 0.8 1/h. (Natural ventilation allowance for a living room.)
A19. Conventional BS EN ISO 6946 surface resistances (R_{si} 0.13 horizontal, 0.10 upward, 0.17 downward; R_{se} 0.04 outside, or the same as R_{si} where the element faces another space; m²K/W) are used where blank, for: Front wall, Gable wall, Party wall. (Small conventional set; enter your own where the construction needs other values.)

4. Working

[8] $R_1 = R_{si} + \Sigma\, d/\lambda + R_g + R_{se}$ [1]
= Front wall: 0.13 + (0.12202 + 3.3333 + 0.5 + 0.0325) + 0 + 0.04
= 4.1579 m²·K/W
Front wall: total thermal resistance

[9] $U_1 = 1/R$ [1]
= 1 / 4.1579
= 0.24051 W/(m²·K)
Front wall: U-value

[10] $\Phi_1 = A \cdot U \cdot f \cdot (T_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 8.7 × 0.24051 × 1 × (21 - -3) °C
= 0.050218 kW
Front wall: fabric loss to θ_e = -3 °C
[11] U_2 = U override (user-entered)
= Window: 1.4 W/m²K
= 1.4 W/(m²·K)
Window: U-value as entered (ground floor or glazing)
[12] $\Phi_2 = A \cdot U \cdot f \cdot (T_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 2.1 × 1.4 × 1 × (21 - -3) °C
= 0.07056 kW
Window: fabric loss to θ_e = -3 °C
[13] $R_3 = R_{si} + \Sigma\, d/\lambda + R_g + R_{se}$ [1]
= Gable wall: 0.13 + (0.12202 + 3.3333 + 0.5 + 0.0325) + 0 + 0.04
= 4.1579 m²·K/W
Gable wall: total thermal resistance
[14] $U_3 = 1/R$ [1]
= 1 / 4.1579
= 0.24051 W/(m²·K)
Gable wall: U-value
[15] $\Phi_3 = A \cdot U \cdot f \cdot (T_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 8.64 × 0.24051 × 1 × (21 - -3) °C
= 0.049872 kW
Gable wall: fabric loss to θ_e = -3 °C
[16] $R_4 = R_{si} + \Sigma\, d/\lambda + R_g + R_{se}$ [1]
= Party wall: 0.13 + (0.0325 + 0.16667 + 0.16667 + 0.0325) + 0.18 + 0.13
= 0.83833 m²·K/W
Party wall: total thermal resistance
[17] $U_4 = 1/R$ [1]
= 1 / 0.83833
= 1.1928 W/(m²·K)
Party wall: U-value
[18] $\Phi_4 = A \cdot U \cdot f \cdot (T_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 8.64 × 1.1928 × 1 × (21 - 18) °C
= 0.030918 kW
Party wall: fabric loss to θ_{adj} = 18 °C
[19] U_5 = U override (user-entered)
= Ground floor: 0.25 W/m²K
= 0.25 W/(m²·K)
Ground floor: U-value as entered (ground floor or glazing)
[20] $\Phi_5 = A \cdot U \cdot f \cdot (T_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 16.2 × 0.25 × 1 × (21 - -3) °C
= 0.0972 kW
Ground floor: fabric loss to θ_e = -3 °C
[21] $A_{fab} = \Sigma\, A$
= 44.28 m²
= 44.28 m²
Total fabric area
[22] A_{ext} = Σ A facing outside
= 35.64 m²
= 35.64 m²
Fabric area facing outside

[23] A_{tb} = Σ A·f facing outside [2](cl. 6.3.2)
= 35.64 m²
= 35.64 m²
Fabric area facing outside weighted by the temperature factor (thermal-bridge allowance applies)
[24] U_m = Σ A·U / Σ A
= 21.467 / 44.28
= 0.48479 W/(m²·K)
Area-weighted mean U-value of the fabric (without the temperature factor f)
[25] U_{eff} = Σ A·U·f / Σ A
= 21.467 / 44.28
= 0.48479 W/(m²·K)
Heat-loss weighted mean U-value of the fabric (includes the temperature factor f)
[26] dT = T_{int} - θ_e
= 21 degC - (-3 degC)
= 24 K
Design temperature difference, inside to outside
[27] Φ_{el} = Σ Φ_i [2](cl. 6.3.2)
= 298.77 W
= 0.29877 kW
Fabric loss through the elements
[28] Φ_{tb} = y_{tb} * A_{tb} * dT [2](cl. 6.3.2)
= 0.1 W/(m²·K) × 35.64 m² × 24 K
= 0.085536 kW
Thermal-bridge allowance
[29] Φ_T = Φ_{el} + Φ_{tb} [2](cl. 6.3.2)
= 0.29877 kW + 0.085536 kW
= 0.3843 kW
Fabric heat loss
[30] q_v = n * V
= 0.8 1/h × 38.88 m³
= 31.104 m³/h
Ventilation air flow
[31] dT_v = dT
= 24 K
= 24 K
Ventilation temperature difference, inside to outside
[32] Φ_V = 0.34 Wh/(m^3*K) * q_v * dT_v [2](cl. 6.3.3)
= 0.34 Wh/(m^3*K) × 31.104 m³/h × 24 K
= 0.25381 kW
Ventilation heat loss (0.34 Wh/m³K = conventional ρ·c_p of air, derived)
[33] Φ_{HL} = Φ_T + Φ_V [2](cl. 6.3.2)
= 0.3843 kW + 0.25381 kW
= 0.63811 kW
Room design heat loss
[34] HLC = Φ_{HL} / dT
= 0.63811 kW / 24 K
= 26.588 W/K
Heat loss coefficient
[36] Φ_A = Φ_{HL} / A_{floor}
= 0.63811 kW / 16.2 m²
= 39.39 W/m²
Room design heat loss per floor area

[37] $U_{target,1}$ = target U of the element
= the global target U_{target}
= 0.3 W/(m²·K)
Front wall: target U-value
[38] $u_{target,1}$ = U / U_{target}
= 0.24051 / 0.3
= 0.8017
Front wall: utilisation of the target U-value
[39] $U_{target,3}$ = target U of the element
= the global target U_{target}
= 0.3 W/(m²·K)
Gable wall: target U-value
[40] $u_{target,3}$ = U / U_{target}
= 0.24051 / 0.3
= 0.8017
Gable wall: utilisation of the target U-value
[41] $U_{target,4}$ = target U of the element
= the target entered on the element's row
= 1.2 W/(m²·K)
Party wall: target U-value
[42] $u_{target,4}$ = U / U_{target}
= 1.1928 / 1.2
= 0.99404
Party wall: utilisation of the target U-value

5. Results

SYMBOL	VALUE	DESCRIPTION
Φ_{HL}	0.6381 kW	Room design heat loss
T_{int}	21 °C	Internal design temperature
HLC	26.59 W/K	Heat loss coefficient
Φ_T	0.3843 kW	Fabric heat loss
Φ_V	0.2538 kW	Ventilation heat loss
U_1	0.2405 W/(m²·K)	Front wall: U-value
R_1	4.158 m²·K/W	Front wall: thermal resistance
Φ_1	0.05022 kW	Front wall: fabric loss
U_2	1.400 W/(m²·K)	Window: U-value
Φ_2	0.07056 kW	Window: fabric loss
R_3	4.158 m²·K/W	Gable wall: thermal resistance
U_3	0.2405 W/(m²·K)	Gable wall: U-value
Φ_3	0.04987 kW	Gable wall: fabric loss
R_4	0.8383 m²·K/W	Party wall: thermal resistance
U_4	1.193 W/(m²·K)	Party wall: U-value (governing)
Φ_4	0.03092 kW	Party wall: fabric loss
U_5	0.2500 W/(m²·K)	Ground floor: U-value
Φ_5	0.09720 kW	Ground floor: fabric loss

SYMBOL	VALUE	DESCRIPTION
A_{fab}	44.28 m²	Total fabric area
A_{ext}	35.64 m²	Fabric area facing outside
A_{tb}	35.64 m²	Fabric area facing outside weighted by f
U_m	0.4848 W/(m²·K)	Mean fabric U-value (area-weighted)
U_{eff}	0.4848 W/(m²·K)	Heat-loss weighted fabric U-value (with f)
dT	24 K	Design temperature difference
Φ_{el}	0.2988 kW	Element fabric loss
Φ_{tb}	0.08554 kW	Thermal-bridge allowance
q_v	31.10 m³/h	Ventilation air flow
Φ_A	39.39 W/m²	Heat loss per floor area
$u_{target,1}$	0.80	Front wall: U against target
$u_{target,3}$	0.80	Gable wall: U against target
$u_{target,4}$	0.99	Party wall: U against target

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	U-value of Party wall against its target	1.193 W/(m²·K)	1.200 W/(m²·K)	0.99	
PASS	U-value of Front wall against its target	0.2405 W/(m²·K)	0.3000 W/(m²·K)	0.80	
PASS	U-value of Gable wall against its target	0.2405 W/(m²·K)	0.3000 W/(m²·K)	0.80	

7. Applicability

WARNING records: tabulated values not taken from a reference record are θ_e (External design temperature); layers (Layer thermal conductivity); y_{tb} (Thermal bridging allowance); T_{int} (Internal design temperatures); n (Air change rates). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
f_{add}: not entered, so the default of 0 % is used.

8. Verification

Template bs.room-heat-loss@1.6.0 Content hash 3868dcf1a33269a1 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Building Components and Building Elements—Thermal Resistance and Thermal Transmittance—Calculation Methods. BS EN ISO 6946. BSI.
2. BS EN 12831-1:2017.
3. CIBSE, Domestic Heating Design Guide.